

Summarization

GLIMPSE: Pragmatically Informative

Multi-Document Summarization for Scholarly

Reviews

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Abstract—As society moves forward, it increasingly values time efficiency and is therefore less interested in reading long texts. Consequently, summarization algorithms are being widely researched to address this issue. In the context of scientific peer review, Glimpse, a summarization method designed to provide a concise yet comprehensive overview of scholarly reviews, is proposed. This method introduces novel uniqueness scores based on the Rational Speech Act framework to identify relevant sentences in reviews. In this report, we reproduce the results claimed in the paper and take a step further by expanding the current model’s application domain to book reviews on Amazon. The scoring algorithm of the model also utilizes a language model that we have changed to see performance results. The results in the end are as expected.

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I. PROBLEM STATEMENT

A. Expected input

1) Dataset: The data is collected from the ICLR conference, which provides open access to reviews and meta-reviews for all submissions through OpenReview². This dataset contains 28,062 reviews for 8,428 submissions to the ICLR conference from 2017 to 2021. Each paper has its own unique ID and a corresponding meta-review. The meta-review is used as the “Gold” review, which serves as the reference against which the generated summaries will be compared.

2) Models: The model operates in two different settings: In extractive settings, candidate summaries are derived by extracting sentences from reviews. In abstractive settings, PEGASUS (Zhang et al. 2019) or BART (Lewis et al. 2020) are utilized to generate the summaries.

B. Addressed Task

The increasing number of paper submissions to conferences has placed a significant strain on the reviewing process. This surge imposes a burden on area chairs, who must carefully read an ever-growing volume of reviews and discern each

reviewer’s main arguments as part of their decision-making process. The GLIMPSE model aims to address this challenge by summarizing reviews and scoring them based on their common and unique perspectives. The highest-scoring summaries are then selected.

In this work, we introduce one key contribution: The original domain of the model, which was focused on paper reviews, has been expanded to include Amazon user reviews on books.

C. Expected output

The expected output consists of four sets of different summaries for each paper/book. These summaries result from a combination of two distinct summarization techniques (Extractive and Abstractive) paired with two separate scoring algorithms (RSA-Speaker and RSA-Listener).

II. METHODOLOGY

A. Data Usage

: The dataset contains columns that are irrelevant to the task at hand. Therefore, only the following columns were retained:

ID: Used to match reviews belonging to the same document.
User’s Review: The main textual data entering our pipeline.
Meta-Review: Considered the “Gold” summary, serving as the evaluation metric for assessing the model’s accuracy. The data was checked for missing values in the target columns and cleaned accordingly. The “Gold” column had approximately 498 missing values. Initially, we attempted to fill these gaps by referencing the “Gold” column of records with the same ID. However, it was discovered that all instances of missing “Gold” values within the same ID were entirely null, making the “Gold” column irreproducible.

B. Candidate Generation

Candidate summaries are generated using two approaches: Extractive Method: Sentences are directly extracted from the source reviews. Abstractive Method: User reviews are summarized using BART (Lewis et al. 2020 (Abstractive method)).

¹Project GitHub repository: https://github.com/your_username/your_repository

C. Informativeness and Uniqueness Scores

The scoring system is based on the Rational Speech Act (RSA) framework, a model of human communication in which both the speaker and listener assume rationality and continue communicating until they reach a common understanding.

Mathematically, this communication process is represented as a utility function that both the speaker and listener seek to maximize. The utility of the communication from the Listener’s point of view is:

$$V_L(s, d) = \log L(d|s) - \text{Cost}(d),$$

where $L(d|s)$ denotes the conditional probability where $L(d|s)$ denotes the conditional probability of guessing the object d — the document/review in our case — upon receiving s — a short summary— according to the listener L , $\text{Cost}(s)$ is the cost of transmitting the summary s .

$$L_0(d|s) = \frac{\text{LM}(s|d)}{\sum_{d \in \mathcal{O}} \text{LM}(s|d)}.$$

As previously mentioned two different scoring algorithms are proposed: 1) Pragmatic-speaker-based-score: This score identifies the most discriminative utterance that refers to a source document. Mathematically, for a given document d , the speaker’s distribution over the summaries is defined as the softmax of the utility scores with a listener L_{t-1} , where L_0 is the literal listener. Pragmatic reasoning is modeled as an iterative process that begins with a literal listener—i.e., a listener who has no assumptions about the speaker. We denote this as $L_0(d|s)$ and define it using a pre-trained language model LM , which, based on the code, is PEGASUS (Zhang et al. 2019):

$$S_t(s|d) = \frac{\exp(V_{L_{t-1}}(d, s))}{\sum_{s'} \exp(V_{L_{t-1}}(d, s'))}.$$

The pragmatic speaker adapts to a listener under a cost constraint by maximizing communication utility (Zaslavsky, Hu, and Levy 2021). It directly scores the summaries for a document according to the pragmatic speaker’s distribution.

In our case, this process evaluates the informativeness of a summary within the set of candidates. Hence, we define the pragmatic-speaker-based score as the argmax of the speaker’s distribution:

$$\text{RSA-Speaker}(d) \triangleq \arg \max_{s \in C} S_t(s|d).$$

This score is referred to as GLIMPSE-speaker in the results. 1) Uniqueness score: This score is defined based on quantifying the listener’s uncertainty regarding a review. If a sentence is given to a listener, the probability with which it selects the correct review represents the uniqueness score, as it is directly related to the uniqueness of that sentence. They propose measuring this uncertainty through:

$$\text{Unique} \triangleq D_{\text{KL}}(L(\cdot|s) \parallel \mathcal{U}).$$

The higher this value would be, the more unique the sentence would be. The model in which implements this scoring is referred to as GLIMPSE-Unique.

III. EVALUATION

The evaluation is conducted on different aspects of the generated results, which are as follows: 1) This evaluation assumes that the meta-reviewer’s decisions provide a reasonable comparison and uses them as the reference in ROUGE score calculations. It utilizes unigram and bigram overlaps, as well as ROUGE-L, which measures the longest common subsequence between the summarized reviews and the gold standard. However, as the paper mentions, ROUGE score is not a fully reliable metric since it has a low correlation with human judgment (Kryscinski et al. 2020). 2) Discriminativity: Based on (Ou, Krojer, and Fried 2023) an evaluation is constructed using the cosine similarity between the summary and the reviews’ paraphrase embeddings (Reimers and Gurevych 2019). 3) SEAHORSE Metrics: The generated summaries are evaluated using SEAHORSE, a summarization evaluation metric trained on human judgment (Clark et al. 2023). The summaries are assessed based on six parameters: Coverage and attribution measure the degree to which summaries match the gold standard in terms of main ideas and the accuracy of information attribution. Comprehensibility and grammar assess language quality, focusing on fluency and grammatical accuracy. Repetition and conciseness evaluate redundant sentences and the overall conciseness of the summary. Among these parameters, Coverage best reflects the effectiveness of the algorithms used in this paper in capturing important aspects of reviews. The results of the evaluation is as seen in table I GLIMPSE reproduced results

IV. EXPERIMENTS

We propose two different extensions to the original paper:

Applying the model to a new domain. Changing the language model responsible for scoring extracted/generated sentences. (Note: We are not referring to the model that generates the summary in the abstractive setting.)

A. Data Description

The dataset consists of Amazon user reviews for books. Our goal was to create a CSV file with a structure similar to that used in the original paper to minimize the effort required for adapting the code.

To achieve this, we structured the CSV file so that each book’s ID appears multiple times—each instance corresponding to a different review of the same book. We then created an additional file that aggregates all reviews for each book.

For the creation of the “gold” column, we concatenated all the reviews from the previous column, added a prompt, and used ChatGPT to generate a summary for each specific book. We selected GPT-4o for this task to ensure that the language in the generated summaries closely resembled that of state-of-the-art language models.

The results of GPT-4o were manually assessed on twenty different book reviews (which we had already read), and the generated summaries were found to be good representations of the original content. In this approach, we consider:

TABLE I
GLIMPSE REPRODUCTION RESULTS

Method	Common Rouge-1	Common Rouge-2	Common Rouge-L
extractive_speaker	0.196512 (0.007276)	0.02637 (0.001047)	0.110223 (0.002475)
extractive_unique	0.26408 (0.006619)	0.03255 (0.001280)	0.12370 (0.002320)
abstractive_speaker	0.303320 (0.007986)	0.065009 (0.002234)	0.162360 (0.002256)
abstractive_unique	0.324837 (0.008049)	0.065235 (0.002160)	0.162230 (0.002187)

Books as analogous to research papers Reviews as equivalent to reviewers’ opinions Gold summaries as a refined synthesis generated with the help of ChatGPT.

B. Experimental Design:

The base model employed for the scoring task is PEGASUS. However, since this is a text summarization task, we replaced PEGASUS with the FLAN-T5 model to evaluate performance differences.

Given that FLAN-T5 comes in various sizes—and considering that most book reviews are relatively short—we selected two small variants: FLAN-T5 Base and FLAN-T5 Large. This allowed us to analyze the impact of model size on the performance of large language models in this task.

C. Results and analysis of the results:

The result of following experiments is as follows: he findings presented in II indicate that applying the model to the new domain of book reviews has resulted in similar or even better performance compared to its original domain. (Assuming that the summaries generated by GPT are accepted as valid.)

On the other hand, modifying the scoring language model does not improve performance. This is expected, as the original language model was specifically fine-tuned for the target task. Interestingly, a larger FLAN-T5 model performed worse in generating results similar to GPT’s, despite expectations that a larger model would behave more similarly to GPT.

V. CONCLUSIONS

In conclusion, we successfully reproduced the model and its functionality, with slight differences in the results. This was followed by a successful extension of the paper into a completely new domain. In other words, the model performed adequately on Amazon book reviews.

Additionally, we experimented with modifying the language model used in the scoring algorithms. The results remained similar to those of the original model, albeit with a slight decrease in performance.

A. Remarkable Outcomes

Unexpectedly, the model performs well not only in its original domain but also in book reviews. This suggests that the model has potential capabilities for summarizing content based on its reviews, indicating a possible application in multi-document summarization tasks.

B. Main Challenges

This model is highly reliant on human judgment, which means that expanding its domains requires a significant amount of human annotation. This posed a major challenge in domain extension, as there are few datasets available that contain their own “Gold” summaries.

Another major challenge is the high computational cost. Some datasets required more than 7 hours to process on A100 GPUs. Furthermore, evaluating these models remains a persistent challenge, as a low ROUGE score from summarizing reviews and comparing them to the Gold summary does not necessarily indicate poor model performance.

C. Lesson Learned

Summarization does not always require direct implementation of language models. Instead, language models can be integrated into algorithms to generate specific results. Domain expansion is challenging due to the need for manual human annotations, as finding a target dataset that perfectly aligns with our algorithm is difficult. A larger language model does not always guarantee better results. Increasing the number of parameters does not necessarily lead to improved performance and, in some cases, may produce unexpectedly poorer results due to the model’s increased complexity and positioning within the language space.

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TABLE II
BOOK DATASET RESULTS

Method	Common Rouge-1	Common Rouge-2	Common Rouge-L
abstractive_speaker	0.346376 (0.004633)	0.083165 (0.002202)	0.184715 (0.002209)
abstractive_unique	0.350142 (0.004671)	0.084335 (0.002197)	0.184795 (0.002257)
extractive_speaker	0.280699 (0.008301)	0.055983 (0.001964)	0.151719 (0.002252)
extractive_unique	0.291914 (0.007449)	0.059937 (0.001985)	0.155628 (0.002059)
FlanT5Base_speaker	0.280699 (0.008301)	0.055983 (0.001964)	0.151719 (0.002252)
flanT5Base_unique	0.291914 (0.007449)	0.059937 (0.001985)	0.155628 (0.002059)
flanT5Large_speaker	0.277700 (0.008173)	0.054973 (0.001951)	0.151044 (0.002339)
flanT5Large_unique	0.284322 (0.007591)	0.058067 (0.001853)	0.154023 (0.002183)

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